

Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

- A. Autoimmune hemolytic anemia
- B. Microangiopathic hemolytic anemia
- C. Paroxysmal nocturnal hemoglobinuria (PNH)
- D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

<think> Given the patient's symptoms of fever , abdominal pain , severe weakness , jaundice , and pallor , along with the laboratory findings of bite cells and Heinz bodies on the peripheral smear , the most likely diagnosis can be deduced . The presence of bite cells and Heinz bodies is indicative of oxidative stress , which is a hallmark of G6PD deficiency . Additionally , the patient's recent travel to Nigeria and receipt of anti-malaria prophylaxis may have exposed him to oxidative stressors that triggered the G6PD deficiency . Normal G6PD levels rule out other causes of hemolysis . The absence of other laboratory findings that would suggest autoimmune or microangiopathic hemolytic anemia , or PNH , also supports G6PD deficiency as the most likely diagnosis . </think><answer> D . Glucose - 6 - phosphate - dehydrogenase (G6PD) deficiency </answer>

